

DATA PRE-PROCESSING

Experiment 1: Handling Missing Values

Questions

1. Create a dataset containing missing values and identify them.
2. Replace missing numerical values using:
 - a. Mean
 - b. Median
 - c. Mode
3. Compare the effect of each imputation technique.
4. Remove rows and columns containing missing values and analyze the impact on the dataset size.

Experiment 2: Data Normalization and Standardization

Questions

1. Normalize a dataset using Min-Max Scaling.
2. Standardize a dataset using Z-score normalization.
3. Visualize feature distributions before and after scaling.

Experiment 3: Encoding Categorical Data

Questions

1. Convert categorical variables into numerical form using:
 - a. Label Encoding
 - b. One-Hot Encoding
2. Compare the output of both encoding methods- Visualization.
3. Apply encoding to a dataset containing multiple categorical features.

Experiment 4: Handling Outliers

Questions

1. Detect outliers using box plots.
2. Identify outliers using the IQR method.
3. Detect outliers using Z-score analysis.

4. Remove outliers and compare dataset statistics before and after removal.
5. Study the impact of outliers on machine learning model performance.

Experiment 5: Data Visualization for Preprocessing

Questions

1. Visualize feature distributions using histograms.
2. Plot pairwise feature relationships using scatter plots.
3. Generate a correlation heatmap.
4. Identify skewed features from visualizations.
5. Suggest preprocessing techniques based on visual analysis.

Experiment 7: Handling Imbalanced Data

Questions

1. Identify class imbalance in a dataset.
2. Apply random oversampling.
3. Apply random undersampling.
4. Implement SMOTE (Synthetic Minority Oversampling Technique).
5. Compare classification performance before and after balancing.

Experiment 8: Data Splitting and Cross Validation

Questions

1. Split a dataset into training and testing sets.
2. Implement k-Fold Cross Validation.
3. Compare different values of k.
4. Evaluate model stability using cross-validation scores.
5. Explain the importance of validation data.